

# Project 2

## Solar Thermal Energy

REE 401

*Dynamic simulation and techno-economic analysis of a solar thermal water-heating system with storage*

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# Abstract

This project studies a solar thermal water-heating system in which a flat-plate solar collector is coupled to a thermal storage tank, a circulation pump, and an auxiliary electric heater. Building on the collector-only system of Project 1, a storage tank is added so that heat collected during the day can be delivered to the load at any hour. The complete system is modelled and simulated in **TRNSYS** over a one-week (168-hour) period, and the resulting fluid temperatures and heat-transfer rates are analysed. A techno-economic assessment is then carried out to determine whether the investment is worthwhile. Using an installed capital cost of \$1,800, an annual useful energy of 1900 kWh, and a Saudi residential electricity price of 0.05 USD/kWh, the system yields net annual savings of about \$86, a simple payback period of roughly **21 years**, and a 25-year return on investment of about **19%**. A sensitivity analysis shows that the economics improve dramatically at higher electricity tariffs, and the system avoids approximately 1.24 tonnes of CO<sub>2</sub> emissions per year.

# Contents

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>Abstract</b>                                    | <b>i</b>  |
| <b>1 Introduction</b>                              | <b>1</b>  |
| 1.1 Objectives . . . . .                           | 1         |
| <b>2 System Description</b>                        | <b>2</b>  |
| 2.1 Solar Collector . . . . .                      | 2         |
| 2.2 Thermal Storage Tank . . . . .                 | 2         |
| 2.3 Circulation Pump . . . . .                     | 2         |
| 2.4 Auxiliary Heater . . . . .                     | 2         |
| 2.5 Controller, Weather Data and Load . . . . .    | 3         |
| <b>3 Methodology: TRNSYS Model</b>                 | <b>4</b>  |
| <b>4 Results and Discussion</b>                    | <b>6</b>  |
| <b>5 Economic Analysis</b>                         | <b>9</b>  |
| 5.1 Capital Cost . . . . .                         | 9         |
| 5.2 Assumptions . . . . .                          | 9         |
| 5.3 Calculations . . . . .                         | 10        |
| 5.4 Sensitivity to the Electricity Price . . . . . | 10        |
| <b>6 Environmental Impact</b>                      | <b>12</b> |
| <b>7 Conclusion</b>                                | <b>13</b> |

# List of Figures

- 3.1 TRNSYS model of the solar thermal system, showing the connections between the weather-data reader, pump, solar collector, storage tank (Type 4), auxiliary heater, load (demand) function, plotter and printer. . . . . 4
- 4.1 Raw output of the TRNSYS online plotter for the complete one-week (168 h) simulation, exactly as produced by the software. . . . . 6
- 4.2 Annotated redraw of the same weekly result as Figure 4.1: the collector and auxiliary heat-transfer rates ( $Q_{SC}$ ,  $Q_{AXR}$ ) together with the collector, auxiliary and pump outlet temperatures over 168 h. The arrows identify the physical meaning of each feature. As in the TRNSYS plotter, the temperatures and the heat-transfer rates share the same numeric scale. . . . . 7
- 5.1 The payback period falls steeply as the displaced electricity becomes more valuable. 11

# List of Tables

|     |                                                                  |    |
|-----|------------------------------------------------------------------|----|
| 3.1 | Solar-collector inputs. . . . .                                  | 4  |
| 3.2 | Thermal storage tank (Type 4) inputs. . . . .                    | 5  |
| 3.3 | Auxiliary-heater inputs. . . . .                                 | 5  |
| 5.1 | Breakdown of the installed capital cost. . . . .                 | 9  |
| 5.2 | Economic assumptions. . . . .                                    | 9  |
| 5.3 | Summary of economic results at 0.05 USD/kWh. . . . .             | 10 |
| 5.4 | Sensitivity of payback and ROI to the electricity price. . . . . | 11 |

## Chapter 1

# Introduction

Solar thermal energy is one of the most direct and efficient ways of using the sun's energy: instead of converting sunlight into electricity, a solar thermal system converts it directly into useful heat. Because roughly a third of the energy consumed in a typical building is used for heating water and spaces, solar thermal collectors can displace a large share of conventional fuel or electricity consumption. This makes the technology particularly attractive for a country such as Saudi Arabia, which receives some of the highest levels of solar irradiance in the world and is actively diversifying its energy mix under Vision 2030.

A basic solar thermal system consists of a collector that absorbs solar radiation and transfers the heat to a working fluid. However, solar energy is intermittent: it is only available during daylight hours and varies with weather. To make the system practical, a means of *storing* heat is required so that hot water is available when it is actually demanded — in the evening or early morning — rather than only when the sun is shining. The purpose of this project is therefore to extend the collector-only system studied in Project 1 by adding a thermal storage tank, to simulate the complete system dynamically, and to evaluate both its technical performance and its economic feasibility.

## 1.1 Objectives

The specific objectives of this project are:

- to build a complete solar thermal system — collector, storage tank, pump and auxiliary heater — in the TRNSYS transient-simulation environment;
- to simulate the system over a representative one-week period and analyse the fluid temperatures and heat-transfer rates;
- to examine how the addition of thermal storage improves the availability of hot water compared with the collector-only system of Project 1;
- to perform a techno-economic analysis (capital cost, annual savings, payback period and return on investment); and
- to estimate the environmental benefit of the system in terms of avoided CO<sub>2</sub> emissions.

## Chapter 2

# System Description

The system studied in this project is a closed solar thermal loop that supplies hot water to a load. Its main components and their functions are described below.

## 2.1 Solar Collector

The solar collector is the heart of the system. It absorbs incident solar radiation and transfers the resulting heat to the circulating working fluid (water). The performance of a collector is commonly described by its instantaneous thermal efficiency, which relates the useful energy gain to the solar radiation incident on the collector aperture:

$$\eta = \frac{\dot{Q}_{\text{useful}}}{G A_c}, \quad (2.1)$$

where  $\dot{Q}_{\text{useful}}$  is the useful thermal power delivered to the fluid,  $G$  is the incident solar irradiance and  $A_c$  is the collector area. In this study a single  $1 \text{ m}^2$  flat-plate collector with an assumed optical/thermal efficiency of 0.8 is used.

## 2.2 Thermal Storage Tank

The thermal storage tank (modelled as TRNSYS Type 4) is the new element added relative to Project 1. It stores the hot water produced by the collector during the day and releases it to the load when required, decoupling energy supply from energy demand. The tank is stratified: hotter water accumulates at the top (the “hot side”) while cooler return water enters near the bottom (the “cold side”). A tank of 1000 litre capacity is used, with a hot-side reference temperature of  $45^\circ\text{C}$  and a cold-side temperature of  $15^\circ\text{C}$ .

## 2.3 Circulation Pump

The pump maintains the flow of the working fluid around the solar loop, driving water from the tank through the collector and back. It consumes a small amount of electrical power, which is accounted for in the analysis. The loop flow rate in the model is  $100 \text{ kg/h}$ .

## 2.4 Auxiliary Heater

Because solar energy alone cannot always raise the water to the required delivery temperature — especially on cloudy days or at night — an auxiliary electric heater is placed in series with the load. It “tops up” the water to the set-point temperature of  $60^\circ\text{C}$  whenever the solar-heated

water falls below it, guaranteeing a reliable supply. The auxiliary heater therefore represents the conventional energy that the solar system is designed to displace.

## 2.5 Controller, Weather Data and Load

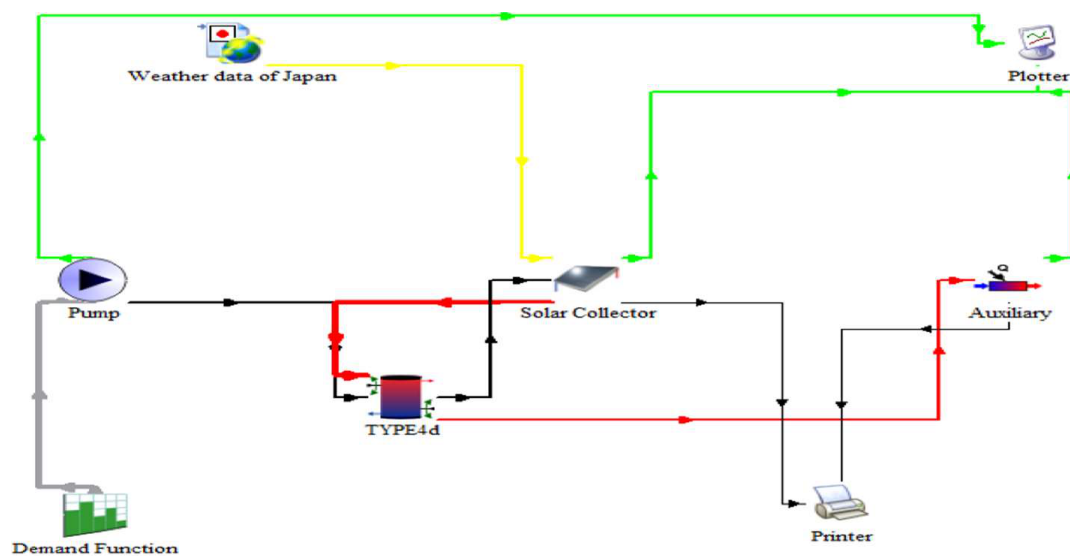
A controller monitors the tank and collector temperatures and switches the pump and heater on or off through control signals. A weather-data reader supplies the hourly solar radiation and ambient temperature that drive the simulation, and a load (demand) function represents the hot-water draw. Plotter and printer components record the results. It should be noted that the weather file used in the model corresponds to **Japan**; for an actual deployment in Ha'il, a Saudi typical-meteorological-year (TMY) file would be substituted, which would increase the collected solar energy owing to the higher local irradiance.



## Chapter 3

# Methodology: TRNSYS Model

The complete system was assembled in TRNSYS, a transient systems-simulation program in which each physical component is represented by a pre-defined module (a “Type”) and the modules are connected so that the output of one becomes the input of the next. Figure 3.1 shows the resulting information flow between the components.



**Figure 3.1.** TRNSYS model of the solar thermal system, showing the connections between the weather-data reader, pump, solar collector, storage tank (Type 4), auxiliary heater, load (demand) function, plotter and printer.

The simulation was run for a total time of 168 h (one week) so that several successive day–night cycles could be observed. The key input parameters assigned to each component are summarised in Tables 3.1–3.3.

**Table 3.1.** Solar-collector inputs.

| Parameter                  | Value            |
|----------------------------|------------------|
| Collector area             | 1 m <sup>2</sup> |
| Optical/thermal efficiency | 0.8              |
| Inlet temperature          | 20 °C            |
| Inlet flow rate            | 100 kg/h         |
| Ambient temperature        | 10 °C            |
| Incidence angle            | 45°              |
| Ground reflectance         | 0.2              |

**Table 3.2.** Thermal storage tank (Type 4) inputs.

| Parameter                 | Value      |
|---------------------------|------------|
| Tank capacity             | 1000 litre |
| Hot-side temperature      | 45 °C      |
| Cold-side temperature     | 15 °C      |
| Environment temperature   | 20 °C      |
| Control signal, element 1 | 1 (on)     |
| Control signal, element 2 | 0 (off)    |

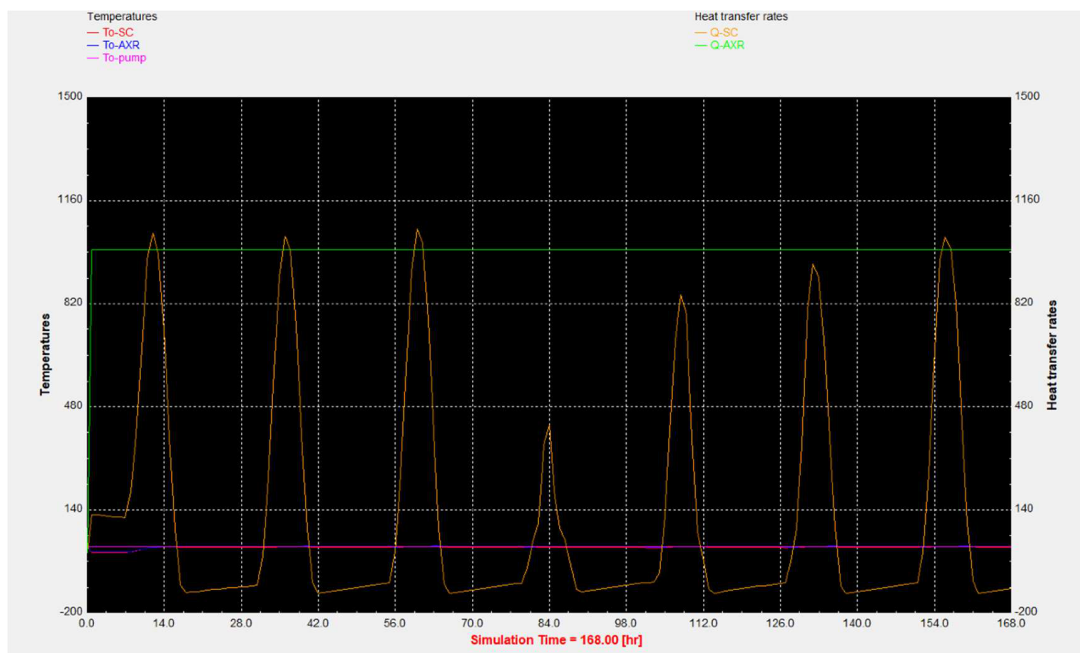
**Table 3.3.** Auxiliary-heater inputs.

| Parameter                   | Value    |
|-----------------------------|----------|
| Inlet fluid temperature     | 20 °C    |
| Fluid mass flow rate        | 100 kg/h |
| Set-point temperature       | 60 °C    |
| Control function            | 1 (on)   |
| Temperature of surroundings | 20 °C    |

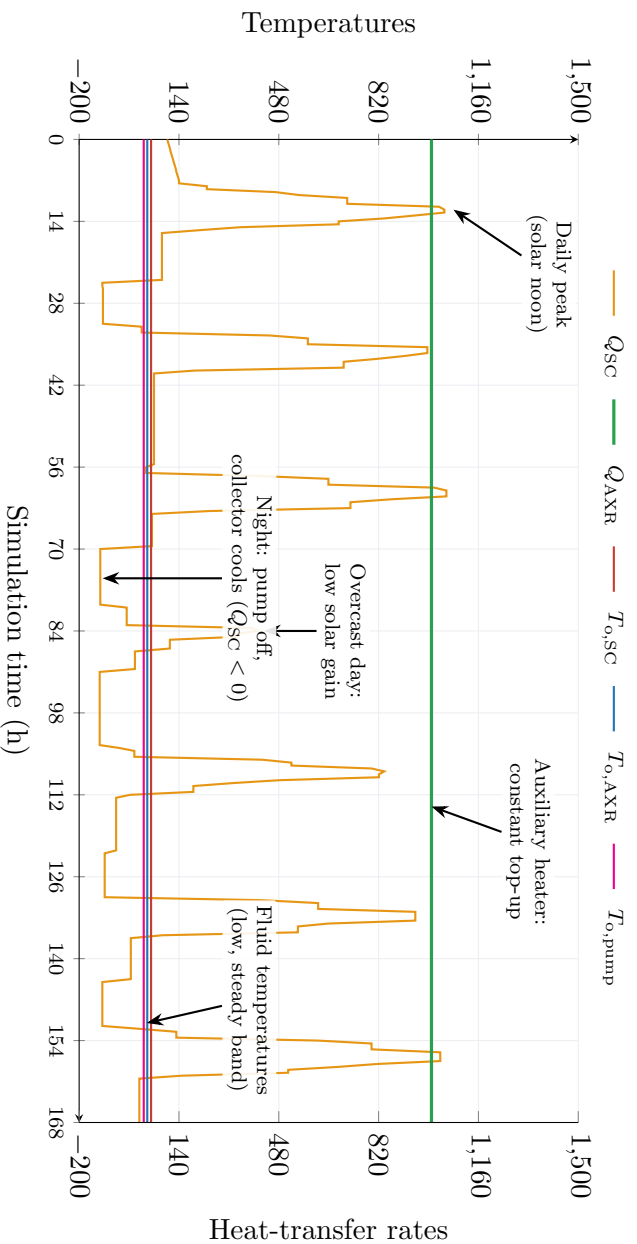
## Chapter 4

# Results and Discussion

The simulation output over the one-week period is presented in two complementary ways. Figure 4.1 reproduces the *raw screen* of the TRNSYS online plotter exactly as the software generates it. Because that screen is dense and hard to read, Figure 4.2 redraws the identical result as a clean, **annotated** diagram in which each important feature is labelled with an arrow. Both figures plot the fluid temperatures (left axis) and the heat-transfer rates (right axis) against time, using the same shared numeric scale.



**Figure 4.1.** Raw output of the TRNSYS online plotter for the complete one-week (168 h) simulation, exactly as produced by the software.



**Figure 4.2.** Annotated redraw of the same weekly result as Figure 4.1: the collector and auxiliary heat-transfer rates ( $Q_{SC}$ ,  $Q_{AXR}$ ) together with the collector, auxiliary and pump outlet temperatures over 168 h. The arrows identify the physical meaning of each feature. As in the TRNSYS plotter, the temperatures and the heat-transfer rates share the same numeric scale.

Several features of the response can be identified. The **collector heat-transfer rate** ( $Q_{SC}$ , orange) shows a clear **daily cycle**: it rises sharply each morning as the sun climbs, reaches a pronounced peak around solar noon, and falls back to near zero at night, dipping slightly negative when the collector loses heat to the cooler surroundings while the pump is off. Seven such peaks are visible, one for each day of the simulated week. Their height varies with the available solar radiation: most days peak at roughly 1,000 units, whereas the fourth day reaches only about 400, indicating an overcast, low-irradiance day.

The **auxiliary heat-transfer rate** ( $Q_{AXR}$ , green) remains almost constant throughout the week. It represents the “top-up” energy that the electric heater must supply to keep the delivery temperature at the 60 °C set point whenever the solar contribution is insufficient — most noticeably at night and in the early morning. Because this line stays high and flat while the solar contribution swings up and down, the figure shows clearly how much of the load still depends on conventional energy.

The three temperature signals — the collector outlet ( $T_{o,SC}$ ), the auxiliary outlet ( $T_{o,AXR}$ ) and the pump outlet ( $T_{o,pump}$ ) — appear as a narrow band near the bottom of the plot. They stay low and nearly flat only because they share the same numeric axis as the much larger heat-transfer rates; on their own scale they vary within a modest range. The low, steady pump-outlet temperature confirms that relatively cool water is returned from the bottom of the tank to the collector inlet, which is desirable because a low inlet temperature keeps the collector efficiency high.

Overall, the results demonstrate that the system behaves as intended: the collector harvests solar energy during the day, the storage tank retains part of it and smooths the supply, and the auxiliary heater guarantees a continuous supply of hot water at the set point. Compared

with the collector-only system of Project 1, the addition of storage reduces the number of hours during which the load would otherwise be left without solar-heated water.

**What the figure actually means.** In essence, the graph is a picture of how the **free solar supply** and the **paid back-up supply** share the heating duty over one week. The seven orange pulses ( $Q_{SC}$ ) are the solar energy captured by the collector: tall pulses are sunny days and the short fourth pulse is a cloudy day. The flat green line ( $Q_{AXR}$ ) is the electrical energy that the auxiliary heater must keep adding so that the water always reaches the 60 °C set point. The relative size of the orange pulses compared with the green line is, in effect, the *solar fraction* of the system: the larger and more frequent the solar pulses, the less the auxiliary heater has to run, and the greater the fuel and cost savings quantified in the economic analysis of the next chapter. The steady low temperature band, finally, confirms that despite the large swings in the *rate* of heat transfer, the delivered water is held within a narrow, usable temperature range — which is exactly the purpose of adding the storage tank.

## Chapter 5

# Economic Analysis

The most important question for any renewable-energy investment is whether it pays for itself. This chapter estimates the capital cost of the system, the annual savings it generates, and the resulting payback period and return on investment.

## 5.1 Capital Cost

The installed capital cost is the sum of the equipment, labour and installation items listed in Table 5.1.

**Table 5.1.** Breakdown of the installed capital cost.

| Item                                               | Cost (USD)   |
|----------------------------------------------------|--------------|
| Storage tank (1000 litre)                          | 500          |
| Circulation pump                                   | 100          |
| Solar collector ( $1 \text{ m}^2$ , $\eta = 0.8$ ) | 500          |
| Auxiliary heater                                   | 200          |
| Labour                                             | 100          |
| Insulation and piping                              | 400          |
| <b>Total initial cost <math>C_0</math></b>         | <b>1 800</b> |

## 5.2 Assumptions

The economic calculation uses the following assumptions, summarised in Table 5.2. Maintenance is taken as 0.5 % of the capital cost per year. The annual useful energy delivered to the load is estimated as the gross collected energy of 2000 kWh multiplied by a storage/delivery efficiency of 0.95, giving 1900 kWh/yr. The value of this energy is set by the Saudi residential electricity price that it displaces, 0.05 USD/kWh.

**Table 5.2.** Economic assumptions.

| Quantity                            | Value        |
|-------------------------------------|--------------|
| Initial capital cost, $C_0$         | \$1 800      |
| System lifetime, $N$                | 25 years     |
| Annual maintenance (0.5% of $C_0$ ) | \$9 / yr     |
| Annual useful energy, $E$           | 1900 kWh/yr  |
| Electricity price, $p$              | 0.05 USD/kWh |

### 5.3 Calculations

The annual gross saving is the value of the electricity displaced by the solar system:

$$S = E \times p = 1900 \text{ kWh/yr} \times 0.05 \text{ USD/kWh} = \$95/\text{yr}. \quad (5.1)$$

Subtracting the annual maintenance cost gives the net annual saving:

$$S_{\text{net}} = S - \text{O\&M} = 95 - 9 = \$86/\text{yr}. \quad (5.2)$$

The simple payback period is the time required for the net savings to recover the initial cost:

$$\text{Payback} = \frac{C_0}{S_{\text{net}}} = \frac{1800}{86} \approx \mathbf{20.9 \text{ years}}. \quad (5.3)$$

Over the 25-year life, the total net saving and the net profit are

$$S_{\text{life}} = N S_{\text{net}} = 25 \times 86 = \$2\,150, \quad \text{Net profit} = S_{\text{life}} - C_0 = 2150 - 1800 = \$350, \quad (5.4)$$

so that the return on investment is

$$\text{ROI} = \frac{\text{Net profit}}{C_0} \times 100 = \frac{350}{1800} \times 100 \approx \mathbf{19.4\%}. \quad (5.5)$$

The results are collected in Table 5.3.

**Table 5.3.** Summary of economic results at 0.05 USD/kWh.

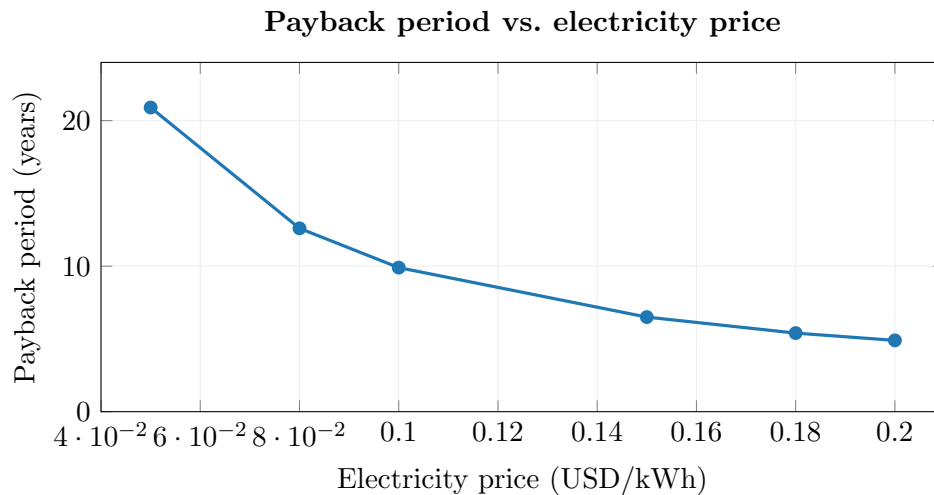
| Metric                      | Value      |
|-----------------------------|------------|
| Annual gross saving         | \$95 / yr  |
| Net annual saving           | \$86 / yr  |
| Simple payback period       | 20.9 years |
| Lifetime net saving (25 yr) | \$2 150    |
| Net profit (25 yr)          | \$350      |
| Return on investment        | 19.4%      |

### 5.4 Sensitivity to the Electricity Price

The single most influential parameter in the analysis is the price of the electricity that the system displaces. At Saudi Arabia's heavily subsidised residential tariff of 0.05 USD/kWh the payback is long, but at the higher tariffs typical of commercial customers or of unsubsidised markets the system becomes very attractive. Table 5.4 and Figure 5.1 show how the payback period shortens as the electricity price rises.

**Table 5.4.** Sensitivity of payback and ROI to the electricity price.

| Price (USD/kWh) | Net saving (\$/yr) | Payback (yr) | 25-yr ROI |
|-----------------|--------------------|--------------|-----------|
| 0.05            | 86                 | 20.9         | 19%       |
| 0.08            | 143                | 12.6         | 99%       |
| 0.10            | 181                | 9.9          | 151%      |
| 0.15            | 276                | 6.5          | 283%      |
| 0.18            | 333                | 5.4          | 362%      |
| 0.20            | 371                | 4.9          | 415%      |

**Figure 5.1.** The payback period falls steeply as the displaced electricity becomes more valuable.

At the base-case price the system only just recovers its cost within its lifetime; however, the analysis shows that a modest increase in the tariff, a larger collector area, or the use of Saudi (rather than Japanese) weather data — all of which raise the value or the quantity of the delivered energy — would move the payback well below ten years.



## Chapter 6

# Environmental Impact

Beyond the direct financial return, the system provides an environmental benefit by displacing electricity that would otherwise be generated from fossil fuels. Taking a grid emission factor of about 0.65 kg of CO<sub>2</sub> per kWh — representative of the oil- and gas-dominated Saudi grid — the annual and lifetime emission savings are

$$m_{\text{CO}_2} = E \times \text{EF} = 1900 \text{ kWh/yr} \times 0.65 \text{ kg/kWh} \approx 1235 \text{ kg/yr (1.24 tonnes/yr)}, \quad (6.1)$$

$$m_{\text{CO}_2, 25\text{yr}} \approx 30.9 \text{ tonnes}. \quad (6.2)$$

Over its lifetime the system therefore avoids roughly 31 tonnes of CO<sub>2</sub>, equivalent to the emissions of a passenger car driven well over 100000 km. This benefit is not captured in the payback calculation but is an important part of the overall value of the investment, particularly in the context of Saudi Arabia's commitments to reduce its carbon intensity under Vision 2030.

## Chapter 7

# Conclusion

This project extended the collector-only solar thermal system of Project 1 by adding a 1000 litre thermal storage tank and simulating the complete system — collector, tank, pump and auxiliary heater — in TRNSYS over a one-week period. The simulation confirmed that the system operates as intended: the collector temperature follows a clear daily cycle, the storage tank retains heat between peaks and smooths the supply, and the auxiliary heater guarantees a continuous supply of water at the 60 °C set point.

The techno-economic analysis showed that, at Saudi Arabia's subsidised residential electricity price of 0.05 USD/kWh, the system has an installed cost of \$1,800, net annual savings of about \$86, a simple payback period of roughly 21 years, and a 25-year return on investment of about 19%. At this tariff the margin is thin, but the sensitivity analysis demonstrated that the economics improve very rapidly with the electricity price: at 0.10 USD/kWh the payback falls to about ten years, and at commercial tariffs it drops below six years. In addition, the system avoids approximately 1.24 tonnes of CO<sub>2</sub> per year, or about 31 tonnes over its lifetime.

In conclusion, solar thermal energy with storage is a technically sound and environmentally valuable option whose financial attractiveness depends strongly on the local electricity tariff and on the scale of the installation. For larger systems, higher tariffs, or the higher irradiance available in Ha'il, the investment becomes clearly worthwhile. Beyond the financial return, the transition to renewable energy protects the environment and supports the sustainable development of the Kingdom.

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